

Outputs, Presentation Results, And Defense Notes

This file explains what outputs the project produces, how to present them, what each PNG shows, and how to defend results that are imperfect or simulation-based.

What To Present In A Review

A strong presentation flow:

1. Explain the problem: microgrids need cost-aware, renewable-aware, battery-safe dispatch.
2. Show data pipeline: NASA POWER weather data cleaned and normalized.
3. Show synthetic operational data: load, tariff, battery, PV output.
4. Show LSTM forecasting results: solar and load predicted vs actual.
5. Show PPO training results: reward curve and action patterns.
6. Show dispatch and metrics: cost savings, grid dependency, renewable utilization.
7. Show dashboard: live KPIs, forecast, recommendation, alerts, scenario simulation.
8. Start live stream: demonstrate dynamic telemetry every 1.5 seconds.
9. Run scenarios: peak load, low solar, tariff spike.
10. Discuss limitations and future upgrades.

Core Numeric Outputs

Output	Value
Cleaned NASA rows	52,608
Missing values after cleaning	0
Duplicate timestamps after cleaning	0
Load range	26.383 to 151.913 kW
Tariff values	2.6, 5.6, 9.2 INR/kWh
Battery capacity	180 kWh
Battery safe SoC band	20 to 90 percent
Battery violations	0
Solar LSTM MAE	2.0748 kW
Load LSTM MAE	3.5385 kW
PPO training timesteps	20,000
Seven-day baseline cost	INR 41,165.47
Seven-day optimized cost	INR 36,130.22
Seven-day savings	INR 5,035.25
Savings percent	12.23 percent
Renewable utilization	93.37 percent
Grid dependency	68.58 percent
Battery health score	92.3

Generated PNG Outputs

Data Cleaning And Weather

Plot	Path	What it shows	How to present
Solar trends	data/outputs/plots/solar_trends.png	Seven-day rolling mean of GHI, DNI, and diffuse irradiance.	Shows seasonal and daily solar resource variability.
Temperature trends	data/outputs/plots/temperature_trends.png	Seven-day rolling mean of temperature.	Shows weather conditions that influence cooling load.
Wind trends	data/outputs/plots/wind_trends.png	Seven-day rolling mean of wind speed.	Shows wind as environmental context, not a dispatched generator.

Synthetic Operational Data

Plot	Path	What it shows	How to present
Load curve	data/outputs/plots/load_curve.png	Realistic load demand with morning/evening peaks and seasonality.	Defends synthetic load realism.
Tariff curve	data/outputs/plots/tariff_curve.png	Time-of-use tariff blocks.	Shows deterministic India-style tariff structure.
Battery SoC validation	data/outputs/plots/battery_soc_validation.png	SoC and battery power over time.	Shows battery remains within safe operating limits.

Forecasting

Plot	Path	What it shows	How to present
Solar prediction vs actual	data/outputs/plots/solar_kw_prediction_vs_actual.png	LSTM solar prediction against actual generated solar.	Shows model tracks PV pattern with low MAE.
Load prediction vs actual	data/outputs/plots/load_kw_prediction_vs_actual.png	LSTM load prediction against actual synthetic load.	Shows model captures load peaks and daily shape.
Solar loss curve	data/outputs/plots/solar_kw_loss_curve.png	Training and validation MSE over epochs.	Shows learning stability and no over epochs.

Plot	Path	What it shows	How to present
			obvious runaway training.
Load loss curve	data/outputs/plots/load_kw_loss_curve.png	Training and validation MSE over epochs.	Shows load model convergence.

Reinforcement Learning

Plot	Path	What it shows	How to present
PPO reward curve	data/outputs/plots/ppo_reward_curve.png	Reward progression during training.	Shows the policy learns better dispatch behavior over timesteps.
PPO action patterns	data/outputs/plots/ppo_action_patterns.png	Distribution/pattern of idle, charge, discharge actions.	Shows learned control behavior is not random.

Cost And Sustainability

Plot	Path	What it shows	How to present
Baseline cost optimization	data/outputs/plots/baseline_cost_optimization.png	Baseline vs optimized daily cost.	Main economic result.
Baseline sustainability metrics	data/outputs/plots/baseline_sustainability_metrics.png	Renewable share, grid dependency, self-sufficiency.	Main sustainability result.
API cost optimization	data/outputs/plots/api_cost_optimization.png	Cost plot generated through API metrics.	Shows backend endpoints produce presentable outputs.
API sustainability metrics	data/outputs/plots/api_sustainability_metrics.png	Sustainability metrics through API.	Supports dashboard/backend consistency.

Scenario Outputs

Plot	Path	What it shows
Normal cost	data/outputs/plots/scenario_normal_cost_optimization.png	
Normal sustainability	data/outputs/plots/scenario_normal_sustainability_metrics.png	
Peak load cost	data/outputs/plots/scenario_peak_load_cost_optimization.png	

Plot	Path	What it shows
Peak load sustainability	data/outputs/plots/scenario_peak_load_sustainability_metrics.png	
Low solar cost	data/outputs/plots/scenario_low_solar_cost_optimization.png	
Low solar sustainability	data/outputs/plots/scenario_low_solar_sustainability_metrics.png	
Tariff spike cost	data/outputs/plots/scenario_tariff_spike_cost_optimization.png	
Tariff spike sustainability	data/outputs/plots/scenario_tariff_spike_sustainability_metrics.png	

Presentation use:

- Show that the EMS is not only a static model.
- It can stress-test abnormal conditions.
- It produces cost, alerts, and decisions for each scenario.

EDA Outputs

Plot	Path	What it shows
Correlation heatmap	data/outputs/eda/eda_correlation_heatmap.png	Relationships among weather, PV, load, tariff, and battery signals.
Hourly profile	data/outputs/eda/eda_hourly_profile.png	Average daily behavior by hour.
Monthly profile	data/outputs/eda/eda_monthly_profile.png	Seasonal behavior.
Temperature-load lag	data/outputs/eda/eda_temperature_load_lag.png	Correlation between temperature and load at different lags.
Load duration curve	data/outputs/eda/eda_load_duration_curve.png	How often high load occurs.
Distributions	data/outputs/eda/eda_distributions.png	Distribution of key signals.
Recent battery/grid profile	data/outputs/eda/eda_battery_grid_recent.png	Recent interaction between battery and grid.

How To Present The Dashboard

Best demo path:

1. Open <http://127.0.0.1:5173>.
2. Confirm status is **online**.
3. Click **Start live stream**.
4. Show live records increasing.
5. Explain KPI strip.
6. Show forecast chart.
7. Show dispatch recommendation and reason.
8. Press **Force charge**, then explain SoC behavior.
9. Press **Force discharge**, then explain grid import behavior.
10. Press **Island mode**, then explain grid import is forced to zero.
11. Return to **Auto dispatch**.

12. Run **Peak load**, **Low solar**, and **Tariff spike** scenarios.

Defending Inaccurate Or Imperfect Results

If Forecast Is Not Perfect

Defense:

Forecasting is probabilistic, not exact. The goal is to reduce operational uncertainty enough to support better dispatch. The current MAE values are small relative to the load and PV operating range, and the dashboard still enforces battery safety even if forecast error occurs.

Use numbers:

- Solar MAE: 2.0748 kW.
- Load MAE: 3.5385 kW.
- Battery safety is hard-bounded.

If Cost Savings Are Only 12.23 Percent

Defense:

12.23 percent is inside the expected 10 to 25 percent target range. Higher savings would require larger battery capacity, more aggressive tariff spread, higher PV penetration, or more volatile load. The current result is realistic and not inflated.

If Grid Dependency Is Still 68.58 Percent

Defense:

The PV capacity is 140 kW and the total seven-day load is much larger than the available solar energy. The EMS cannot invent energy. It can only improve when to use battery and renewable energy. Grid dependency remains because this is a grid-connected microgrid, not an oversized islanded system.

If Battery Cycles Look High

Defense:

The dispatch is evaluated over a cost-sensitive window with ToU arbitrage. The health score remains healthy at 92.3, and SoC stays inside 20 to 90 percent. Future work can add stronger degradation-aware reward terms.

If Report Says Continuous PPO But Code Is Discrete

Defense:

The project implements the operator-level action requirement: charge, discharge, and idle. Discrete actions improve explainability and safety for the dashboard. Continuous action PPO is a valid future upgrade and is already documented as a mismatch.

If There Is No Real Live Data

Defense:

Real SCADA access is not available in the project environment. The backend live simulator is a digital-twin style telemetry generator that preserves physical constraints. The architecture is ready to replace the simulator with real meter, inverter, and BMS data sources.

Strongest Claims To Make

Use these in presentation:

- The project is not only a forecasting notebook. It is a full-stack EMS.
- It includes data cleaning, ML, RL, dispatch, APIs, dashboard, scenarios, alerts, and MATLAB export.
- Battery safety is enforced by code, not just shown in slides.
- The dashboard explains decisions in operator language.
- The live simulator makes the system dynamic without requiring real SCADA hardware.
- Cost savings are realistic and within expected range.

Claims To Avoid Or Qualify

Avoid saying:

- "This is ready for direct real-grid deployment."
- "The PPO action space is continuous."
- "Wind power is dispatched."
- "The dashboard uses real live SCADA data."
- "MATLAB Simulink physical model is complete."

Say instead:

- "This is a production-style prototype."
- "The current PPO action space is discrete for operator clarity."
- "Wind speed is used as a weather feature."
- "Live data is simulated through a backend telemetry generator."
- "MATLAB export files are generated for Simulink validation workflow."